

What is the `size_t` data type in C?

`size_t` is an unsigned integral data type which is defined in various header files such as:

```
<stddef.h>, <stdio.h>, <stdlib.h>
```

It's a type which is used to represent the size of objects in bytes and is therefore used as the return type by the **sizeof operator**. It is guaranteed to be big enough to contain the size of the biggest object the host system can handle. Basically the maximum permissible size is dependent on the compiler; if the compiler is 32 bit then it is simply a typedef (i.e., alias) for **unsigned int** but if the compiler is 64 bit then it would be a typedef for **unsigned long long**. The `size_t` data type is never negative. Therefore many C library functions like *malloc*, *memcpy* and *strlen* declare their arguments and return type as `size_t`. For instance,